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API-driven Cloud Native Solutions

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CC ZG506 / SE ZG506

Lecture No. 4

Contact Session wise Coverage

Session	Coverage	
CS01	API Basics (OpenAPI, REST, gRPC, GraphQL)	} Basics
CS02	API Basics + Cloud Native Application	
CS03	Cloud Native Application	
CS04	Introduction to Data Science & Machine Learning, DataOps & MLOps	} Data Science & ML
CS05	API driven Cloud-native Data Pipeline	
CS06	API-driven ML pipelines	
CS07	Tools review; Case Study	
CS08	Basic APIs for AI Applications - NLP, CV, Speech recognition	} Cognitive AI
CS09	Language Models (LLMs)	
CS10	RAG Models	
CS11	RAG Models	
CS12	Cloud Native Application Deployment	
CS13	IoT and Data Analytics	} IoT
CS14	APIs for IoT and Data Analytics	
CS15	Serverless Functions; Case study	
CS16	Course review and discussion	

Agenda



Session 4:

1. Big Data and Characteristics
2. Introduction to Data Science and Machine Learning
3. Data Science Process

Session 5:

- DataOps
- Building an API driven Cloud-native Data Pipeline



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The Data Story



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Big Data

Data Deluge

innovate

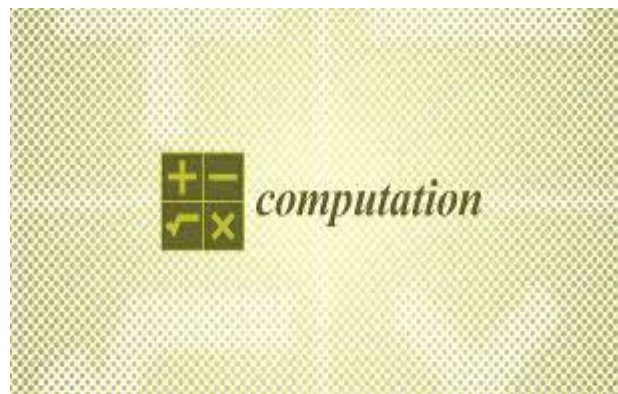
achieve

lead



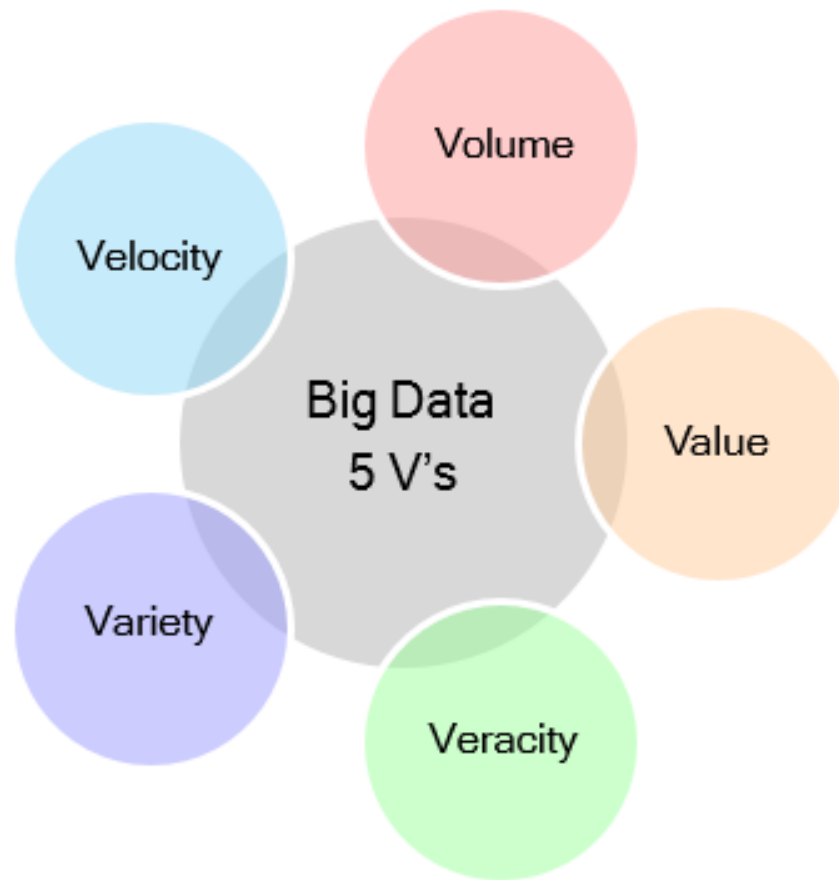
Big Data

- Big Data analytics deals with collection, storage, processing, and analysis of this massive scale data.



Analytics

Characteristics of Big Data



Characteristics of Big Data

1. Volume

- Volume of data involved is so large that it is difficult to store, process and analyze data on a single machine.
- Volumes of data generated by IT / IoT systems is growing exponentially.
- Lowering costs of data storage and processing architectures [possible due to Cloud]
- Need to extract valuable insights from the data to improve business processes, efficiency and service to consumers.

2. Velocity

- Velocity of data refers to how fast the data is generated.
- High velocity of data results in the volume of data accumulated to become very large, in short span of time.

Characteristics of Big Data

3. Variety

- Variety refers to the forms / types of the data.
- Big data comes in different forms such as structured, unstructured or semi-structured, including text data, image, audio, video and sensor data.

4. Veracity

- Veracity refers to how accurate is the data. Need to consider [data provenance](#)
- To extract value from the data, the data needs to be cleaned to remove noise.

5. Value

Value of data refers to the usefulness of data for the intended purpose.

- The value of the data is also related to the veracity or accuracy of the data.
- For some applications value also depends on how fast we are able to process the data.
- Should the data be aggregated to be processed? [Ex: IoT sensors capturing air pollutant parameters]



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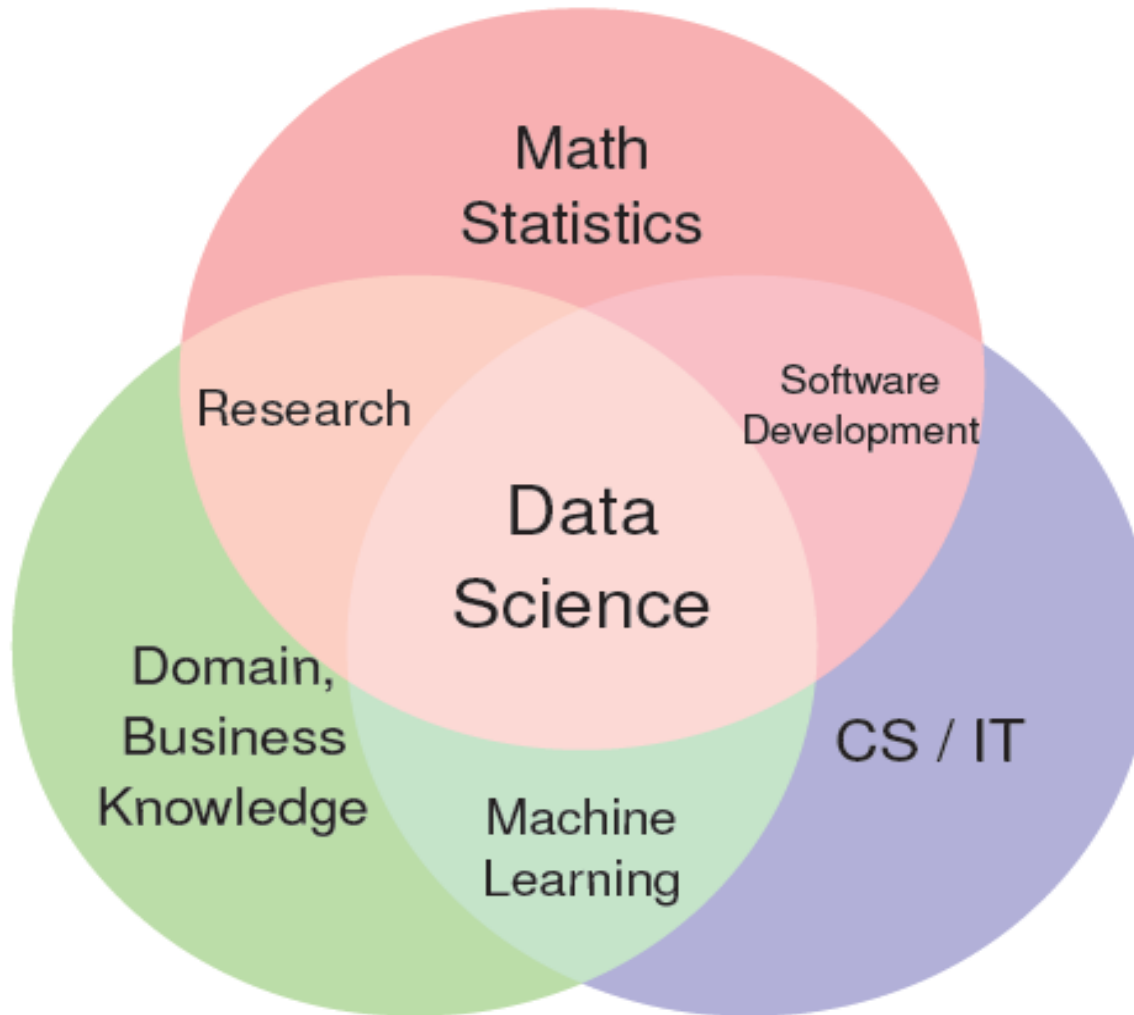


Data Science

Data Science [Some definitions]

- Data Science is the "study of data".
- Data Science is an art of uncovering insights and trends that are hiding behind the data.
- Data Science helps to translate data into a story. The story telling helps in uncovering insights. The insights help in making decision or strategic choices.
- Involves all aspects of “Data” including data collection, preprocessing, analysis, prediction, visualization (storytelling) and gaining insights.

Data Science – Interdisciplinary Field

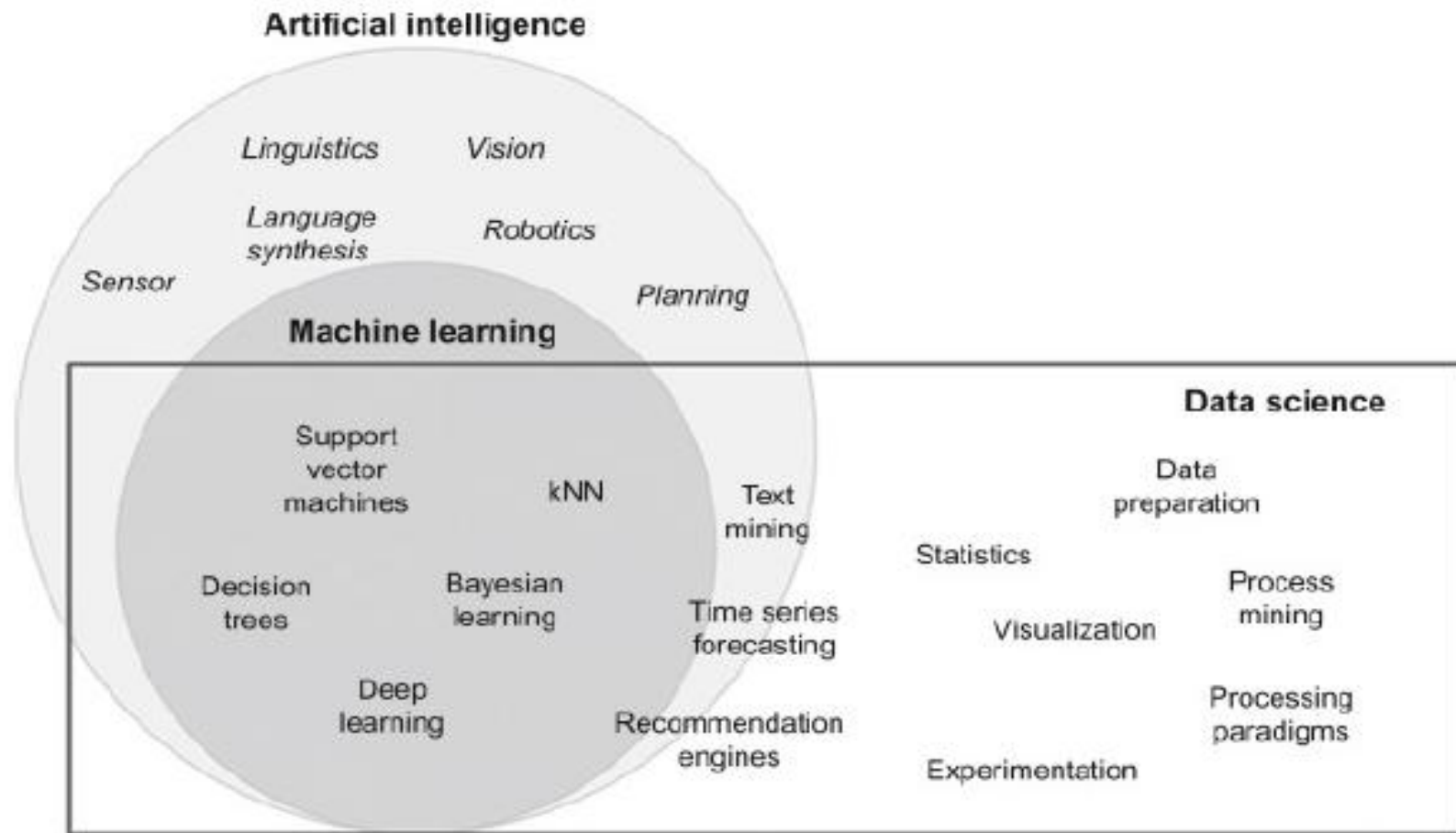


Supportive Technologies for Data Science

Supportive technologies:

1. Powerful algorithms to support computation [Ex: Transformer models like Goole BERT, Open AI's GPT 4 etc.]
2. Open source software and tools [Python]
3. Computational speed, accuracy and cost [Cloud Computing - Azure, AWS, GCP]
4. Data storage in terms of capacity and cost.

Data Science, AI, and ML Convergence

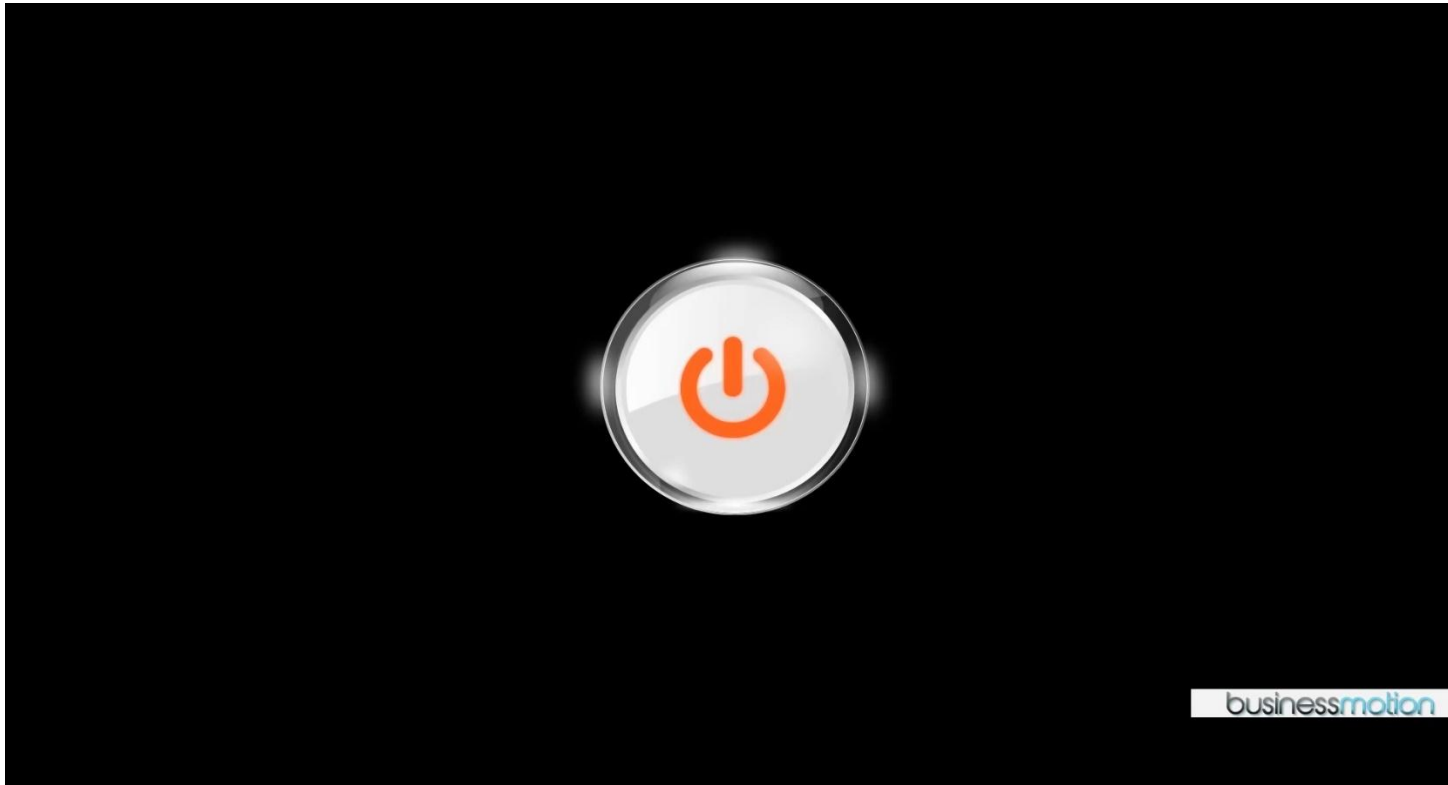


<https://www.sciencedirect.com/topics/physics-and-astronomy/artificial-intelligence>



Roles in Data Science Project

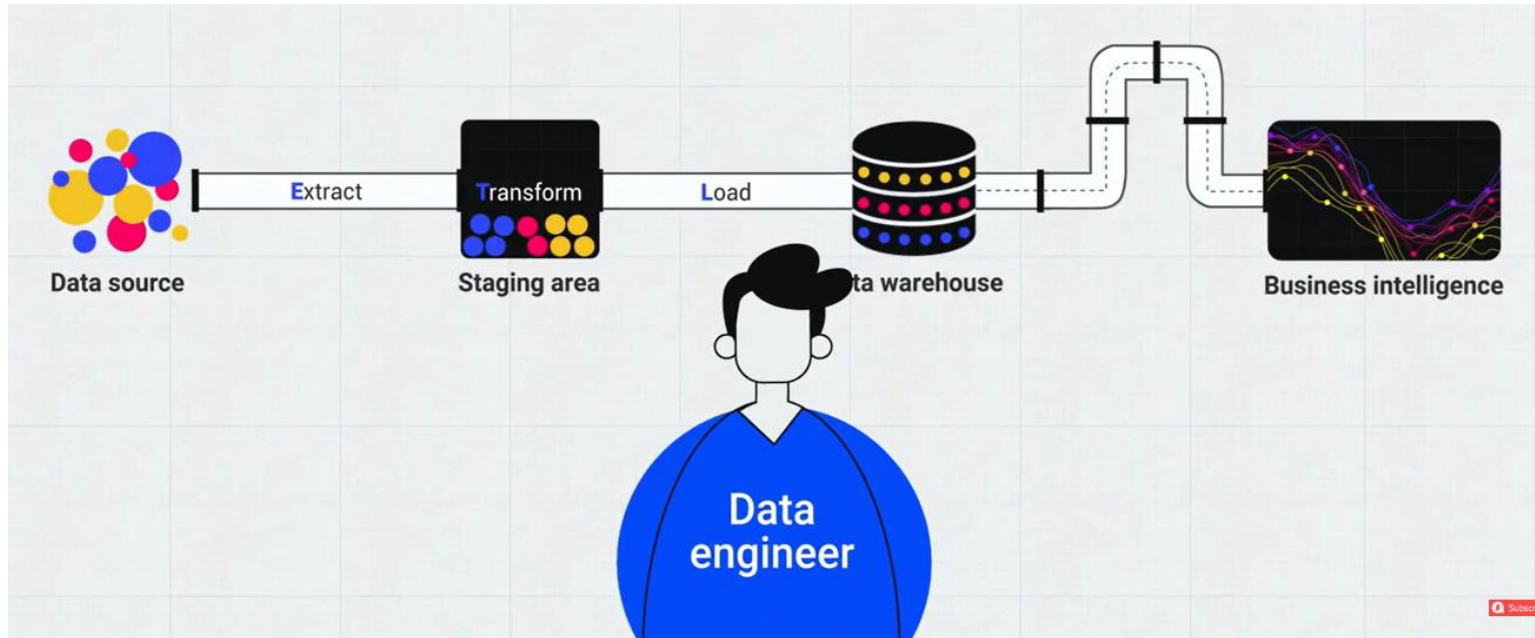
Roles in Data Science Project [10 mins]



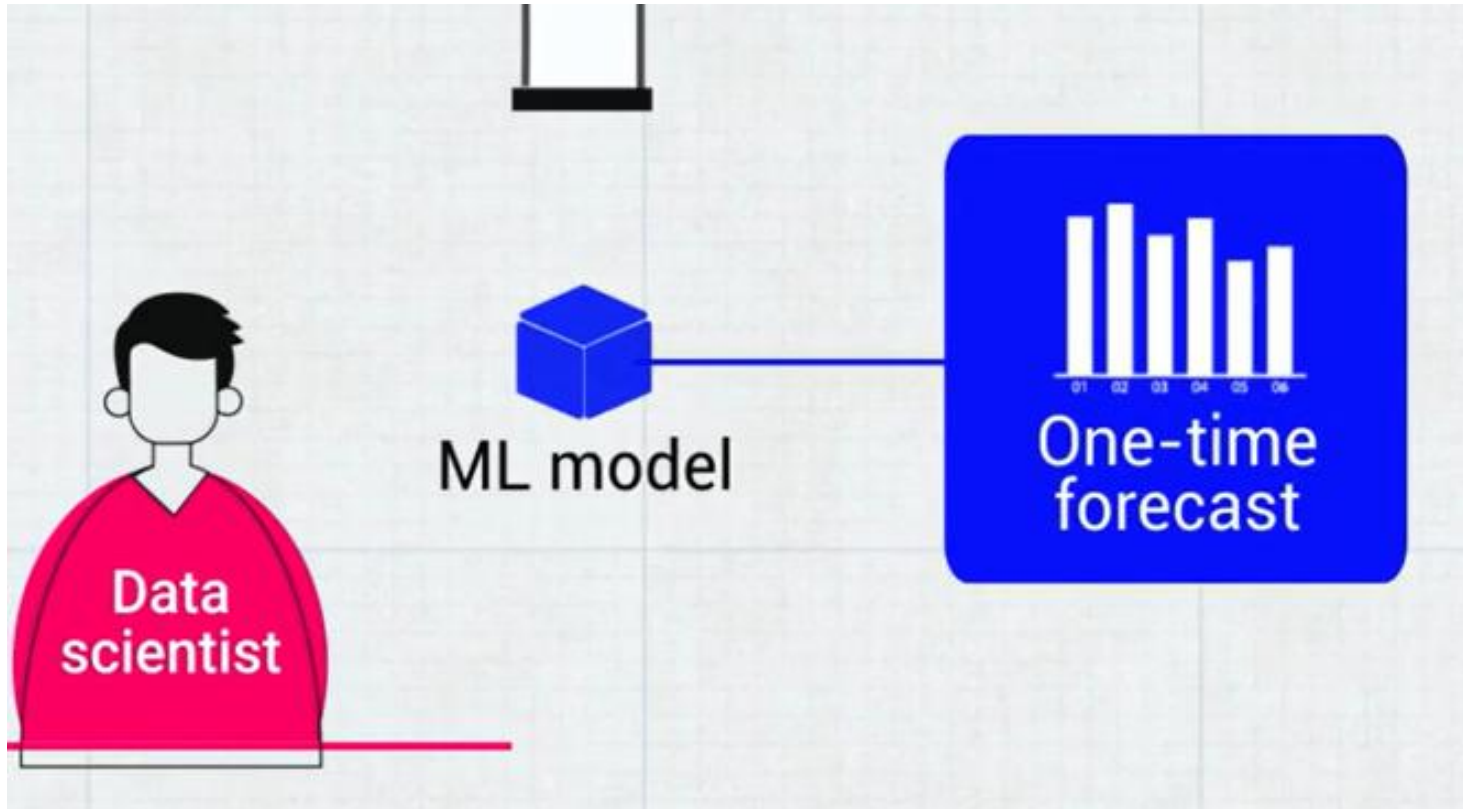
<https://www.youtube.com/watch?v=m5hLUknli5c>

<https://www.altexsoft.com/blog/datascience/how-to-structure-data-science-team-key-models-and-roles/>

Key Takeaways - I



Key Takeaways - II



Key Takeaways - III

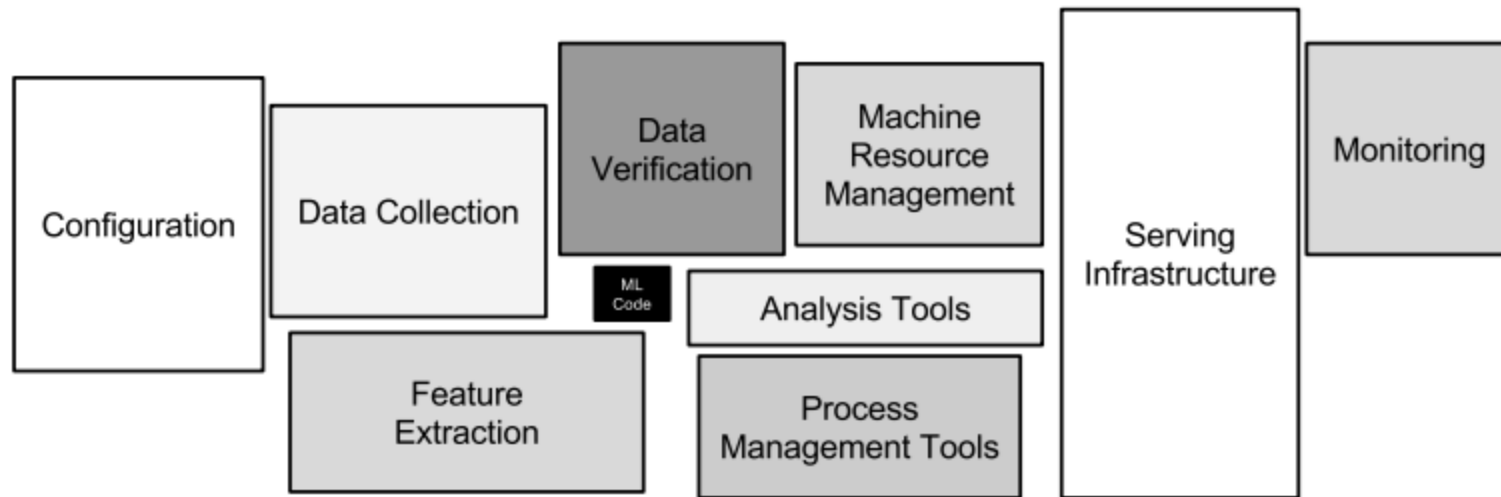
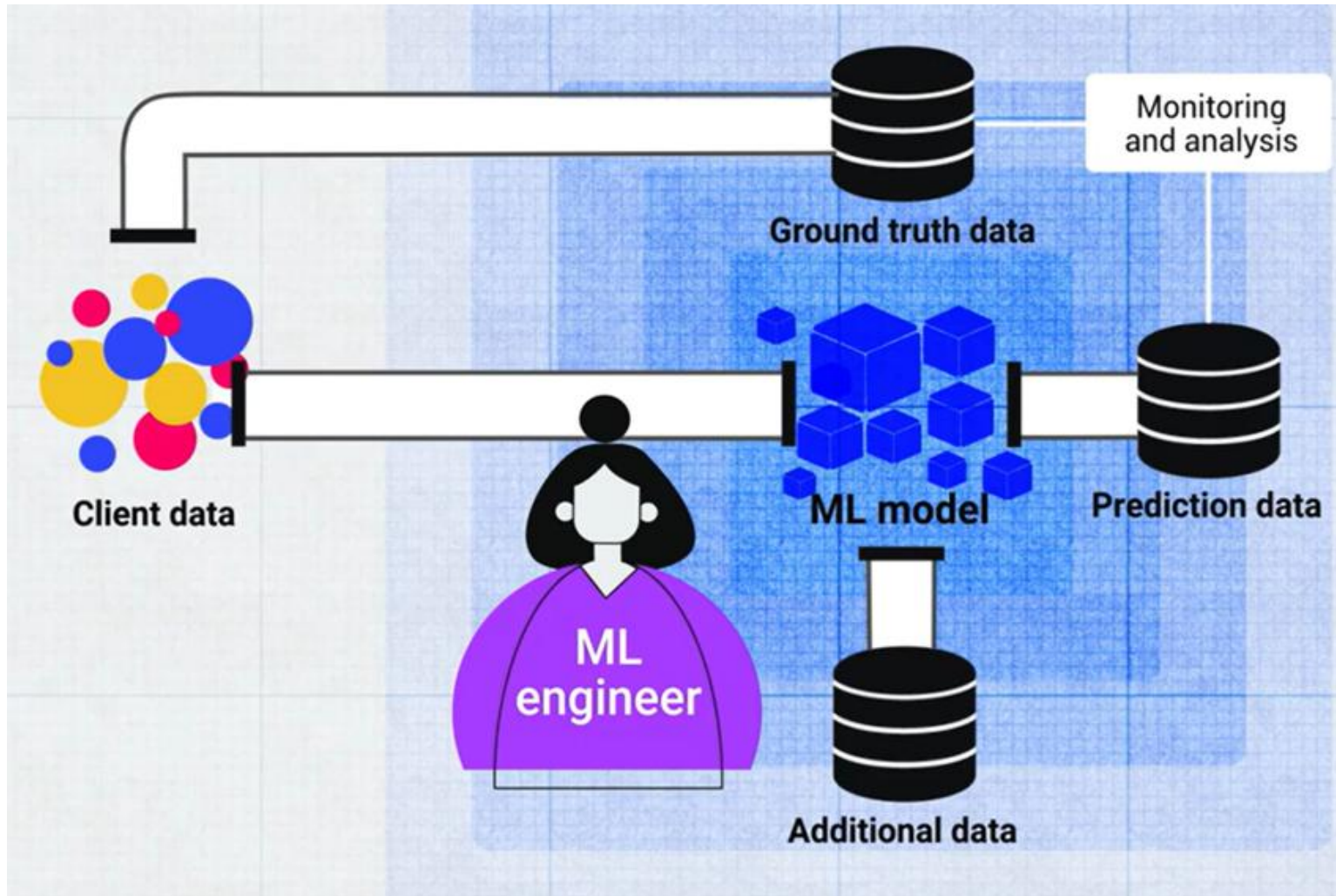


Figure 1: Only a small fraction of real-world ML systems is composed of the ML code, as shown by the small black box in the middle. The required surrounding infrastructure is vast and complex.

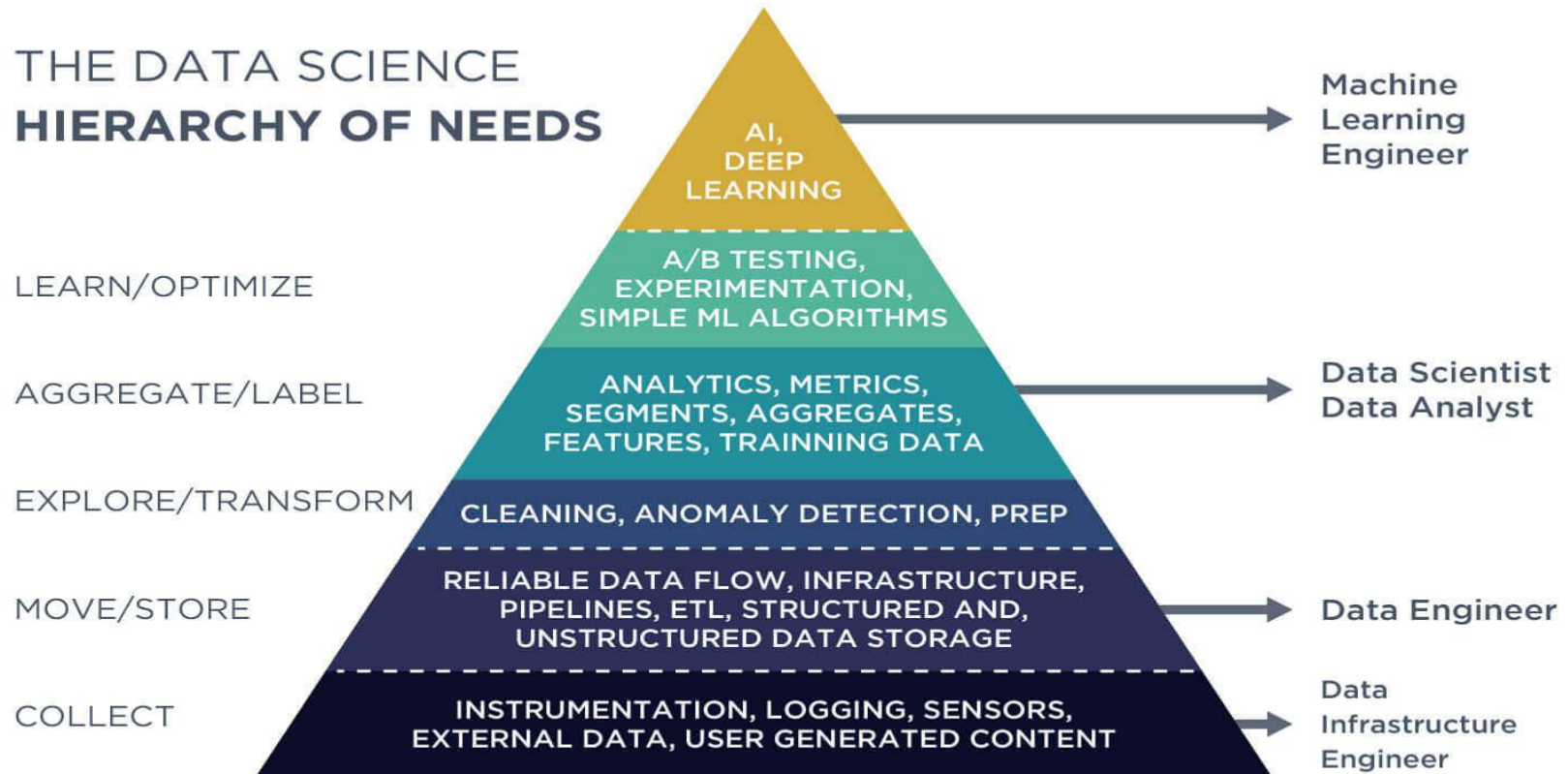
Hidden Technical Debt in Machine Learning Systems

https://proceedings.neurips.cc/paper_files/paper/2015/file/86df7dcfd896fcdf2674f757a2463eba-Paper.pdf

Key Takeaways - IV



Data Science – Hierarchy of Needs



Jelvix

Source: <https://miro.medium.com>

jelvix.com



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Data Science Process / Methodology



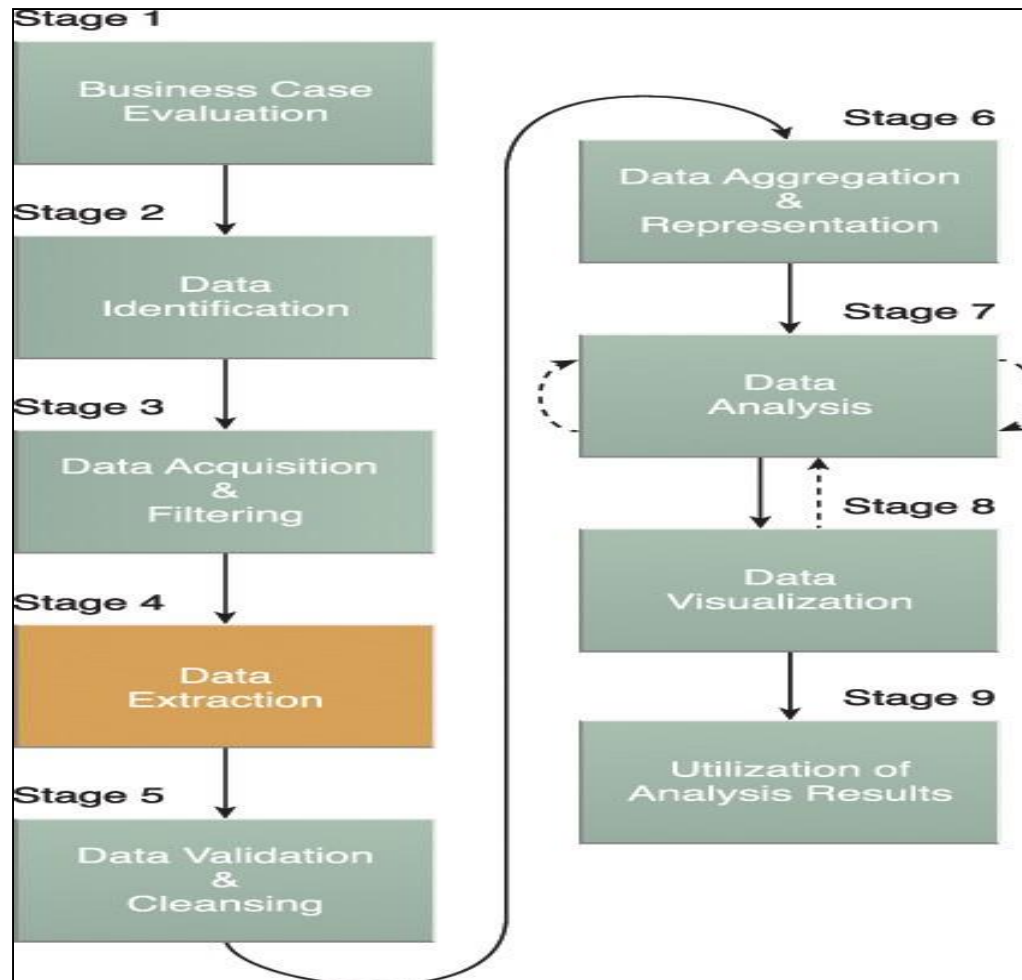
Data Science Process / Methodology

- The data science process (methodology) is a structured approach used to extract valuable insights from data, typically involving a series of iterative steps.
- This process is widely adopted across industries to tackle complex data-driven problems and build predictive models

Some popular Methodologies are:

1. CRISP-DM (Cross Industry Standard Process for Data Mining)
2. DASC-PM (Data Science - Process Model)
3. Big Data Analytics Life Cycle for Data Science projects
4. SEMMA (Sample, Explore, Modify, Model, Assess) applied for ML projects

Big Data Analytics Life Cycle



Case Study



Case: "Predicting Patient Survival After Cardiac Surgery Using Classification Models"

Process: Big Data Analytics Life Cycle

Case Study



Stage 1: Business Case Evaluation

Objective: Define the business goal, which is to predict the survival of patients within 30 days after cardiac surgery. This prediction helps identify high-risk patients who may require additional care and monitoring.

Activities: Stakeholders such as cardiologists, surgeons, and hospital administrators are involved in understanding the clinical and operational benefits of the predictive model. The key performance indicators (KPIs) could include reduction in post-operative mortality rates and improved patient management.

Case Study



Stage 2: Data Identification

Objective: Identify the data sources needed for the analysis.

Activities: The data sources include electronic health records (EHRs) with patient demographics, pre-operative health data (e.g., blood pressure, cholesterol levels), intra-operative data (e.g., surgery duration, anesthesia type), and post-operative data (e.g., complications, ICU stay). Additional data like patient history and lifestyle factors may also be relevant.

Role: Data Architect

Case Study



Stage 3: Data Acquisition and Filtering

Objective: Gather and filter the data necessary for building the predictive model.

Activities: The required data is collected from the hospital's EHR system and other relevant databases. Filtering (at this stage is optional) is performed to ensure data quality, such as removing incomplete records or formatting all the date records into a particular format.

Role: Data Engineer

Case Study



Stage 4: Data Extraction

Objective: Extract the necessary data fields from the collected datasets for further analysis.

Activities: Specific data fields like age, gender, comorbidities, surgery details, and recovery indicators are extracted from the broader datasets. This ensures that only the most relevant information (for the use case) is used for building the model.

Role: Data Engineer

Case Study



Stage 5: Data Validation and Cleansing

Objective: Validate and clean the data to ensure its accuracy and reliability.

Activities: The data undergoes a thorough cleansing process, where missing values are imputed, outliers are identified and treated, and data types are standardized. For example, numerical data like age and surgery duration may be **normalized**, while categorical data like surgery type is **encoded**.

Role: Data Analyst

Case Study



Stage 6: Data Aggregation & Representation

Objective: Aggregate and represent the data in a way that is suitable for analysis.

Activities: The data from multiple sources (e.g., pre-operative, intra-operative, and post-operative data) is aggregated to create a comprehensive dataset for each patient. Features are engineered, such as creating risk scores based on multiple health indicators, which can be used as inputs for the classification models.

Role: Data Architect, Data Scientist

Case Study



Stage 7: Data Analysis

Objective: Perform analysis to build predictive models that classify patient survival.

Activities:

- Perform Exploratory Data Analysis (EDA), which involves univariate, bivariate and multivariate analysis
- Build ML models using classification algorithms like SVM, Random Forest, ANN etc.
- Evaluate the performance of the models, using metrics like Accuracy, Precision, Recall etc.

Role: Data Scientist

Case Study



Stage 8: Data Visualization

Objective: Visualize the results of the data analysis to gain insights and communicate findings.

Activities: Visualization using charts, graphs, dashboards. These visualizations help stakeholders understand the model's performance and the factors most strongly associated with survival. [Ex: Python, Tableau, Power BI etc.]

Role: Data Scientist, Data Visualization Engineer

Case Study



Stage 9: Utilization of Analysis Results

Objective: Apply the results of the analysis in a practical setting to improve decision-making.

Activities: The predictive model is deployed within the hospital's clinical decision support system. Real-time predictions are provided for patients scheduled for cardiac surgery, allowing healthcare providers to identify high-risk individuals and take preventive measures, such as closer monitoring or additional interventions post-surgery.

Role: Machine Learning Engineer, Business Team, Intervention Team / Program Director, Clinical Staff, IT Team



Thank You!